

1. Abstract Securing digital evidence in live forensics, as well as providing court-proof defense against highly complex image manipulations (deepfakes), presents modern investigative authorities with an unsolved architectural problem. Classical cryptographic hashing methods (such as SHA-256) only validate the static final state of a file, but do not allow for spatial, structural verification of volatile memory (RAM). This paper introduces Spatial Forensics, a novel holographic cryptographic architecture. The method combines an asymmetric, deterministically mutating stream cipher ("Avenue Principle") with a three-dimensional collision space ("Parking Garage Principle"). By specifically mixing raw memory addresses and spatially scattering the data streams, a mathematically unique, 3D-verifiable path is generated. Every bit manipulation, no matter how small, leads to an immediate chaos-theoretic collapse of this spatial path (avalanche effect). The system enables a deterministic, court-proof 'Chain of Custody' in which the cryptographic proof is generated directly in-memory and inseparably woven into the structure of the target data.

2. Problem Statement and Limits of Previous Methods (State of the Art) Current digital evidence preservation relies primarily on static, one-dimensional hashing methods. In modern cybercrime fighting, these architectures exhibit two critical vulnerabilities: First, they possess no spatial or structural sensitivity. A classical hash value is purely binary – it collapses completely with the slightest alteration to the file, yet provides no information whatsoever about the structural integrity of the original data. In the case of highly professional deepfakes, these methods fail to spatially isolate the manipulation. Second, the "Chain of Custody" during the infiltration of volatile memories is extremely vulnerable. Since classical forensic signatures are often only generated after network transfer on the investigators' computers, defense attorneys regularly argue that the evidence was falsified during transmission. There is a lack of a system that inseparably validates the memory dump directly on the target hardware.

3. System Architecture: Deterministic 3D Hashing The core of Spatial Forensics forms a high-performance, two-stage pipeline in a native C implementation.

- **3.1. The Asymmetric Mutation Engine ("Avenue Principle"):** Instead of processing memory blocks statically, the algorithm sequentially iterates over the raw data stream. Two cryptographic anchor variables are computed with the respective byte value through an asymmetric base equation. After each read step, the system enforces a deterministic "swap" (the continuous mutation of the anchors). Mathematical bit shifts prevent the variables from scaling into infinity; instead, they form a self-contained, chaos-theoretic attractor.

- **3.2. The Spatial Mapper ("Parking Garage Principle"):** To make the data stream spatially verifiable, the mutated anchor states are passed through a dedicated bit mixer after each iteration. Three-dimensional coordinates (X, Y, Z) are extracted from the resulting 32-bit value. The one-dimensional data stream is thus projected into a predefined spatial grid (e.g., a $32 \times 32 \times 32$ matrix). Targeted, mathematically predictable spatial collisions serve as a signature of structural integrity.

4. Mathematical Properties and Forensic Security

- **4.1. Chaos-Theoretic Stability:** The value range of the anchors is artificially bound, whereby the system generates stable, fractal orbits. In the 3D collision space, this leads to intentional spatial overlaps, the metric of which acts as a secondary fingerprint.
- **4.2. Spatial Avalanche Effect:** If only a single bit is manipulated in a digital piece of evidence (e.g., a photo), a divergent value flows into the asymmetric equation. From exactly this point, the deterministic chain breaks off. The 3D path collapses, allowing the forensic software to spatially isolate the exact starting point of the manipulation.

5. Concrete Application Areas (Use Cases)

- **5.1. Stealth Acquisition (Live Forensics):** Integration as a shared library (.so) into C-based event loop schedulers. This enables data to be cryptographically stamped in real-time directly in the RAM of the target machine, even before it is extracted via the network.
- **5.2. Steganographic Image Sealing:** The generated spatial hologram path is hidden in the least significant bits (LSB) of digital evidence photos. The image invisibly carries its own mathematical 3D structure within itself.

6. Proof of Concept (PoC) The mathematical feasibility was validated in Python and native C. The test scenarios prove that the swap logic generates a stable, reproducible collision signature in a 32-bit grid. The C implementation operates as a self-sufficient, highly dynamic memory module without external dependencies and can be integrated directly into forensic frameworks.